

Week 3, Lecture 2: Bestiary of Distributions

Instructor notes – Part 1: prep reading, Part 2: podium cheat sheet

This single file has two parts, both instructor-only (not a student handout). **Part 1** is read beforehand – full worked examples and the reasoning behind the flow. **Part 2** (after the page break, large print) is the podium copy.

Source material: Bolker 4.4-4.6 (p. 155-188): analyzing probability distributions (moments), the bestiary of distributions, and extending simple distributions. Also mined: `LECTURE2.Rmd` (legacy version of this lecture pair – its Poisson/binomial/beta R demos are reused below with fresh parameter values), and Monday’s lecture notes.

Where this sits in the week

- Monday: probability foundations, Bayes’ rule.
- Tuesday (Lab 3): Act 1 practiced Bayes’ rule; Acts 2-3 got hands-on with `dbinom()`/`dpois()`/`dnorm()`/`pnorm()`/`qnorm()` on a camera-trap monitoring scenario, *before* today’s lecture formally names and organizes these distributions.
- **Today:** name what the lab already touched, add moments and two distributions the lab didn’t cover (a skewed continuous one, and a quick look at what happens when simple distributions don’t fit), and give students a framework for choosing a distribution from the shape of their own data.

Rationale for the flow

Bolker’s own advice for section 4.5: skim it as a map, not a memorization list. Students already have `dbinom()` and `dpois()` in their hands from yesterday’s lab – today isn’t a first introduction, it’s organizing what they already did and adding vocabulary (moments, support, the data-generating-process framing) plus 1-2 new shapes. Keep the survey shallow and wide, with one deeper worked example (moments) and one genuinely new distribution (a bounded continuous one – beta – since nothing in the lab needed it).

Part A: Moments (~8-10 min)

Moments describe the shape of a distribution. The n th moment is $E[X^n]$; the n th *central* moment replaces X with $X - \mu$.

- 0th central moment: always 1 (axiom 2, restated)

- 1st central moment: always 0 (that's what makes μ the mean)
- 2nd central moment: the **variance**
- 3rd, standardized: **skewness**; 4th, standardized: **kurtosis**

Worked derivation (Calc-II algebra only, no new calculus): show that $\text{Var}(X) = E[X^2] - \mu^2$, a formula students will use constantly but rarely see derived.

$$\text{Var}(X) = E[(X - \mu)^2] = E[X^2 - 2\mu X + \mu^2] = E[X^2] - 2\mu E[X] + \mu^2$$

Since $E[X] = \mu$:

$$= E[X^2] - 2\mu^2 + \mu^2 = E[X^2] - \mu^2$$

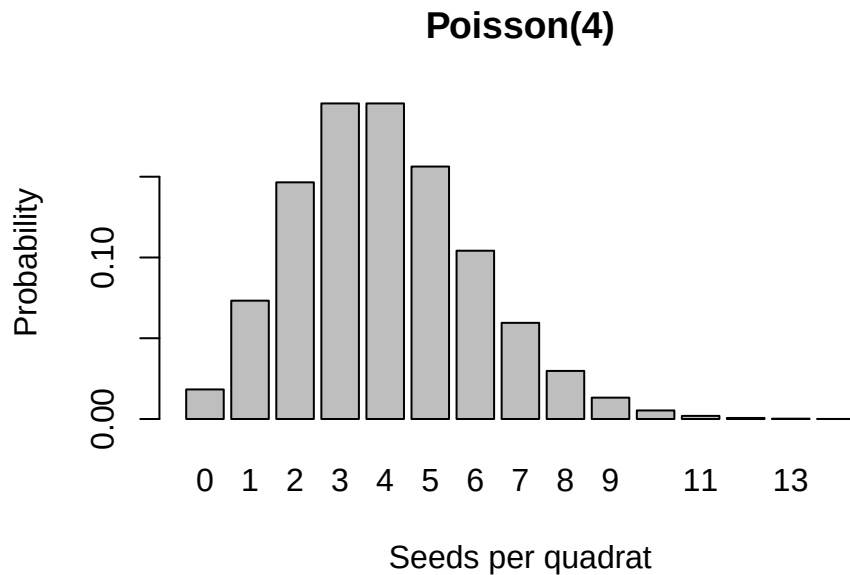
Worth pointing out explicitly: this is just expanding a square and using linearity of expectation – nothing here is harder than what Week 1's algorithms lecture already assumed they could do.

Parameters vs. moments: parameters are the numbers that go into a distribution's formula (e.g. λ for Poisson); moments are properties you can compute *from* the distribution once you know its parameters. Sometimes they're the same thing (Poisson's λ is simultaneously its mean *and* its variance) and sometimes they're not (the normal's σ^2 is its variance directly, but the gamma's variance is a function of two separate parameters).

Part B: Discrete distributions – counts (~8 min)

Fresh example, not yesterday's camera-trap detections: seed counts per quadrat. Suppose the average number of a given plant's seeds found per 1m^2 quadrat is $\lambda = 4$.

```
lambda <- 4
xvals <- 0:14
probs <- dpois(xvals, lambda)
barplot(probs, names.arg = xvals, ylab = "Probability",
        xlab = "Seeds per quadrat", main = "Poisson(4)")
```



```
sum(probs)      # doesn't quite hit 1 -- tail beyond x=14 is still out there
```

```
## [1] 0.9999801
```

Binomial recap (already met in Lab 3 Act 2): counts of successes out of a fixed number of independent trials, each with the same success probability p . **Poisson recap (also met in lab):** counts of independent events in a fixed window of time or space, with rate λ – the limiting case of the binomial as $N \rightarrow \infty$, $p \rightarrow 0$, with $Np = \lambda$ held fixed (this is exactly what Act 2's $N = 100$, $p = 0.024$ comparison demonstrated numerically).

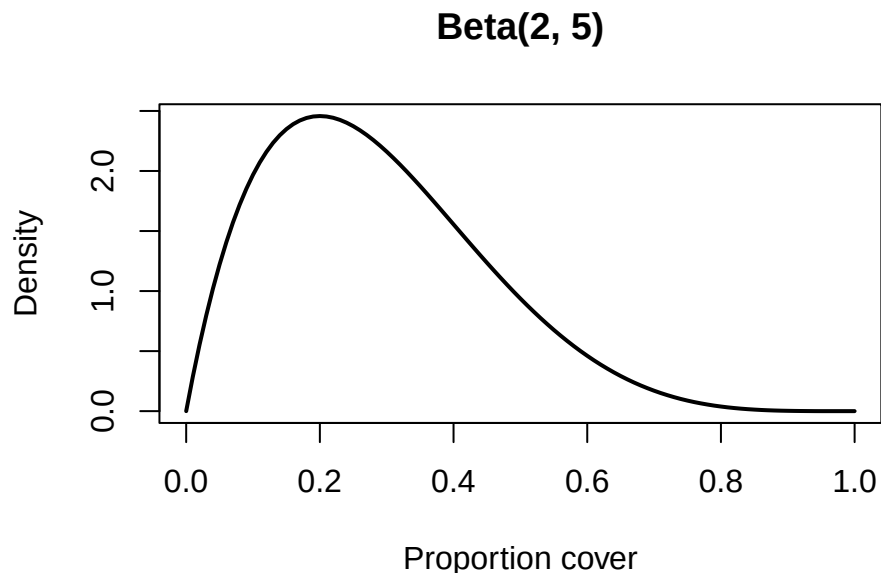
Sticking point to flag: Poisson's single parameter λ is *simultaneously* the mean and the variance (mean = variance is called **equidispersion**). Real count data is very often *more* variable than this (**overdispersion**) – worth a one-sentence preview that the negative binomial exists specifically to relax this constraint (not covered today).

Part C: Continuous distributions – measurements (~8-10 min)

Normal recap (met in Lab 3 Act 3): symmetric, unbounded, fully determined by its first two moments (μ , σ^2) – central to the whole course because of the central limit theorem.

New today – the beta distribution, for data that's a *proportion* or *probability* itself (bounded between 0 and 1). Example: fraction of a quadrat's area covered by an invasive species.

```
a <- 2; b <- 5      # shape parameters
curve(dbeta(x, a, b), from = 0, to = 1, lwd = 2,
      xlab = "Proportion cover", ylab = "Density", main = "Beta(2, 5)")
```



```
integrate(dbeta, lower = 0, upper = 1, shape1 = a, shape2 = b) # check: integrates to
```

```
## 1 with absolute error < 1.1e-14
```

Point out: beta’s two shape parameters (α , β) don’t map onto mean/variance as directly as the normal’s do, but the mean works out to $\alpha/(\alpha + \beta)$ – worth showing since it’s exactly the “fraction of total” intuition the shape parameters suggest. This is also the natural distribution for *a probability itself* – worth a one-sentence flag forward to the Bayesian inference weeks, where a prior on a probability parameter is very often a beta.

Sticking point to flag (recap from Lab 3 Act 3, worth restating with a new example): the height of `dbeta()`’s curve is a *density*, not a probability – it can exceed 1 (try `dbeta(0.2, 2, 5)`), and only an *area* under the curve is a probability.

Part D: Choosing a distribution (~6-8 min)

The organizing question, straight from Bolker: **what process generated this data?**, not “what shape does the histogram look like?” A table worth putting on the board:

Data looks like...	Consider...
successes out of a fixed number of trials	binomial
counts of independent events, no fixed upper bound	Poisson (or negative binomial, if overdispersed)
a bounded proportion or probability	beta
a continuous, roughly symmetric measurement	normal

Data looks like...	Consider...
a continuous, right-skewed, positive measurement	gamma or lognormal (not covered in depth today)

The two “not covered in depth today” mentions (negative binomial, gamma/lognormal) are deliberate – flag them as *named* so students recognize them later in the semester, without deriving them now.

Take-home message

- Moments describe shape; $\text{Var}(X) = E[X^2] - \mu^2$ is worth remembering, not just looking up.
- Distributions are chosen by the *mechanism* that generated the data, not by eyeballing a histogram shape.
- Density $\neq$ probability (only area under a density curve is a probability); Poisson’s mean = variance is a real, checkable assumption, not just a mathematical curiosity.

Roadmap tie-in

- The shapes surveyed today (Weeks 1-2’s deterministic functions) will soon get noise wrapped around them – Week 5 (synthetic data simulation) combines a deterministic mean function with one of today’s distributions to build a full stochastic model.
- Overdispersion (Poisson’s broken equidispersion assumption) comes back explicitly in Week 12 (modeling variance).

Discussion seed (if time remains)

Ask: *looking back at yesterday’s lab – which distribution from today’s tour best describes the camera-trap detection counts (Act 2)? The body mass measurements (Act 3)? Can you imagine a way reality might violate the distribution you picked (e.g., detections clustering in time rather than being independent)?*

PART 2 – PODIUM CHEAT SHEET

Podium copy – terse on purpose. Full derivations, rationale, and code: see Part 1 above.

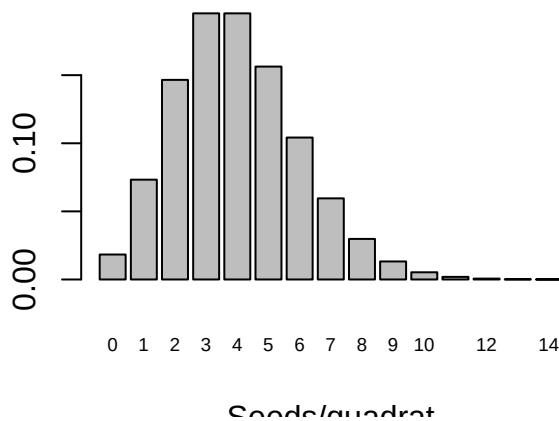
0. Open (~2 min)

- Yesterday’s lab already used `dbinom()`, `dpois()`, `dnorm()` – today organizes and names what you touched, adds moments + one new shape

1. Moments (~8-10 min)

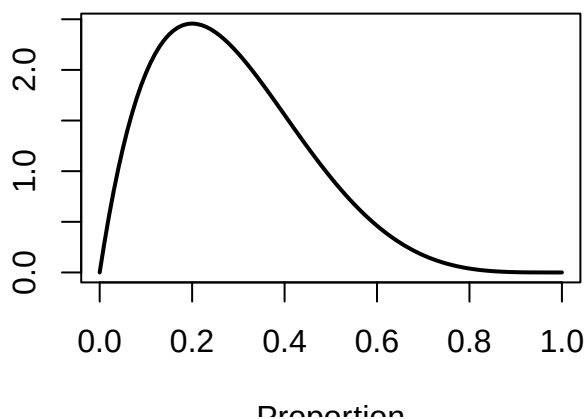
- 0th central moment = 1, 1st = 0, 2nd = **variance**, 3rd = skewness, 4th = kurtosis
- **Derive on board:** $\text{Var}(X) = E[X^2] - 2\mu E[X] + \mu^2 = E[X^2] - \mu^2$ (just expand the square + linearity of $E[\cdot]$)
- parameters (go into the formula) vs. moments (computed from it) – Poisson’s λ is *both* at once

2. Discrete: counts (~8 min)



- **Binomial** (Lab 3 Act 2): fixed N trials, constant p
- **Poisson** (Lab 3 Act 2): rate λ , limit of binomial as $N \rightarrow \infty, p \rightarrow 0, Np = \lambda$
- **Sticking point:** Poisson mean = variance (**equidispersion**) – real counts are often **overdispersed** (negative binomial handles this, not today)

3. Continuous: measurements (~8-10 min)



- **Normal** (Lab 3 Act 3): symmetric, unbounded, μ, σ^2 fully determine it (CLT)
- **New – Beta**: bounded $[0, 1]$, for proportions/probabilities; mean = $\alpha/(\alpha + \beta)$; natural prior for a probability parameter (preview: Bayesian weeks)
- **Sticking point (again)**: density $\neq$ probability – can exceed 1, only *area* is a probability

4. Choosing a distribution (~6-8 min)

- Fixed trials, successes $\rightarrow$ **binomial**
- Counts, no fixed upper bound $\rightarrow$ **Poisson** (or neg. binomial if overdispersed)
- Bounded proportion $\rightarrow$ **beta**
- Symmetric continuous $\rightarrow$ **normal**
- Skewed positive continuous $\rightarrow$ gamma/lognormal (named only)
- **Match the mechanism, not the histogram shape**

Discussion (if time)

- *Which distribution fits yesterday's detection counts (Act 2)? Body mass (Act 3)? How might reality violate that choice?*